

Element B: Documentation & Analysis of Prior Design Attempts

Prior Design Attempt #1: Student Engagement in the Library by Books & Bytes

Source

Student Engagement in the Library blog: www.booksandbytescanada.com/blog/student-engagement-in-the-library

This prior design attempt was found from the Student Engage in the Library blog. This This or That poll was made because the author of the blog found that “Students LOVE sharing their opinions about lots of different topics!” She also found that, “if they have [a This or That poll] in the library, they are likely to visit more frequently.”

Image (source: Books & Bytes blog)



Details

This design uses clear containers, labels with pictures and pom-poms to encourage students to participate in a voting activity. Students place a chosen pom-pom into the container with their chosen option. This activity is visually appealing and easy for students to understand. Also, the materials are also inexpensive and easy to set up.

Analysis

The idea is cute, and I really like the pictures about the option. However, this design is too small. Also, it's in 3 pieces (two containers, and 1 bag of pom-poms). This could lead to it being more likely to be knocked over.

Prior Design Attempt #2:

Source: The Librarian

Image (the librarian gave it to us)



Details:

This design uses 3 clear containers (similar to the prior attempt), 1 per each choice and 1 for the pompom in the middle. Students can choose between two options by placing a pompom into the container that they prefer more.

Analysis:

This design is clear and easy to follow. However, the main critique is that it's too many containers (3 in total). It's easier to knock something over.

Prior Design Attempt #3:

Source: The Librarian

Image (the librarian gave it to us)



Details:

This design is a 3D printed challenge station that lets students vote by dropping pompoms in either option.

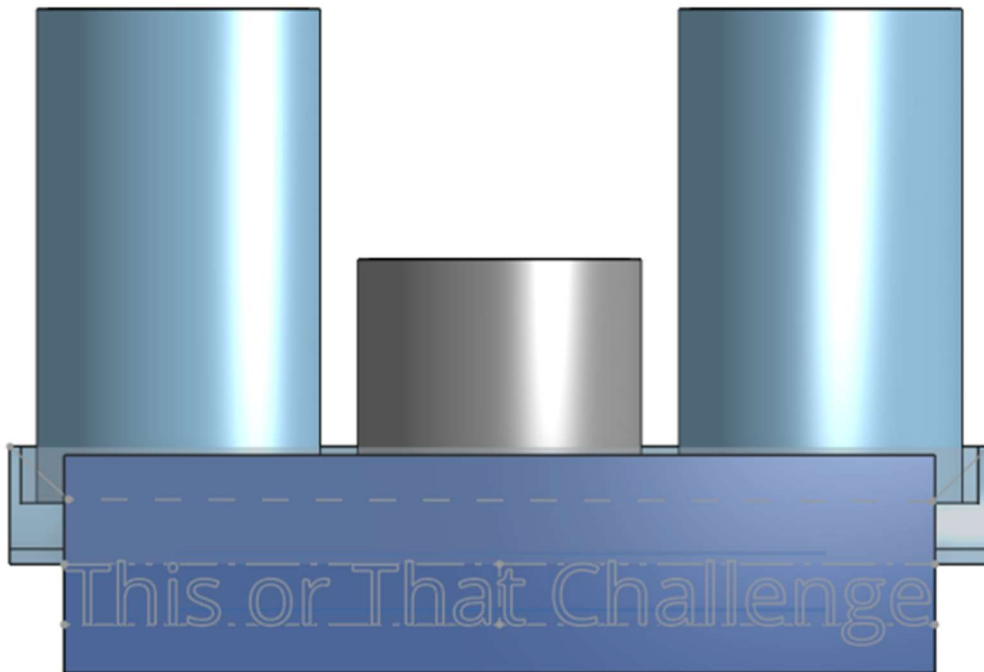
Analysis:

This design is very good in the idea that it's in one piece. Also, the materials is very durable (its made of 3d printer filament). However, the problem is that the cups are too small. There are many students in the school. There should be enough for every student and a bit more space for extra.

Prior Design Attempt #4:

Source: This was our first design

Image: From our CAD (Computer-Aided Design)



Design:

This design is supposed to be 3D printed. It's in one piece. In the design, it has 2 bigger containers (which should be printed using a see-through filament) for each option and one middle container for the pompoms to take.

Analysis:

This design is very easy to understand and build. However, the problem with this design is that there is no place to put each option. Also, it's hard to remove the pompoms when the librarian wants to restart the activity.

Miroboard: miro.com/app/board/uXjVGpkaPUU=

Problem Definition:

A library needs a "This or That" challenge that students can participate in while they wait in line.

Prototype sketch:

First design prototypes:

None of us had a 3D printer, so instead we used commonly found materials such as glass and cardboard. The final design should be made of regular 3D printing filament.

Final iteration prototype:

First design CAD:

Final iteration CAD:

Iteration Changes:

We decided to add Velcro strips on the base and the bottom of the cups. We realized that it's very difficult to empty the cups without having to turn the entire device over so our iteration will allow the cups to be detached and reattached from the base. The second iteration we decided to add question cards on the cups. We realized that it was pretty hard to see which option is each of the cups, so we decided to make it clear what cup is the first option and what cup is the second. These changes still keep our core simple design but add on it.

Testing Plan:

Test #	Test Description	Test Results	Test Status
1	Test the prototype in a line.	1. The prototype is a line.	Pass
2	Test the prototype in a line.	2. The prototype is a line.	Pass
3	Test the prototype in a line.	3. The prototype is a line.	Pass
4	Test the prototype in a line.	4. The prototype is a line.	Pass
5	Test the prototype in a line.	5. The prototype is a line.	Pass
6	Test the prototype in a line.	6. The prototype is a line.	Pass
7	Test the prototype in a line.	7. The prototype is a line.	Pass
8	Test the prototype in a line.	8. The prototype is a line.	Pass
9	Test the prototype in a line.	9. The prototype is a line.	Pass
10	Test the prototype in a line.	10. The prototype is a line.	Pass

Feedback:

Feedback 1: The prototype is a line.

Feedback 2: The prototype is a line.

Feedback 3: The prototype is a line.

Feedback 4: The prototype is a line.

Feedback 5: The prototype is a line.

Feedback 6: The prototype is a line.

Feedback 7: The prototype is a line.

Feedback 8: The prototype is a line.

Feedback 9: The prototype is a line.

Feedback 10: The prototype is a line.

Presentation Video:

<https://docs.google.com/videos/d/14PLoEs9P-lazyfzRce2pkACFoV8MO6pNbCm8wBNU/edit?scene=id.p#scene-id.p>

Most of our tests passed. However, due to the change of materials, a few tests, such as test 4 and 5, did not work.

